

Sasin Gudipati

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Software Engineer specializing in systems performance and developer tooling for simulation and high-reliability systems. Experienced in TypeScript, Python, and Rust with a track record of improving automation, reliability, and simulation fidelity at Boeing and Garmin. Passionate about modernizing legacy systems and building efficient, maintainable software.

Experience

The Boeing Company | Seattle, WA

Full-Stack Software Engineer | February 2024 - Present

- **Modernized flight deck simulation systems** by redesigning React.js (TypeScript) web apps used for pilot training and engineering visualization. Added key features like live simulation refresh, status filtering, and real-time error notifications, improving reliability and usability across multiple programs.
- Served as **SCRUM Master** for the Virtual Flight Deck (VFD) modernization program, owning sprint planning, delivery timelines, and cross-team coordination; **introduced biweekly informal demos to promote knowledge sharing**, surface risks early, and recognize individual contributions, improving alignment across avionics, UI, and simulation teams.
- **Reduced release cycle from 2 days to 30 minutes** by fully automating CI/CD pipelines for Java, C#, and Python tools, standardizing build, test, and deployment processes across engineering applications. **Developed custom automation and tooling** to improve developer efficiency, including a copyright marker, build health monitor, and branch summarizer, enabling faster code reviews and reducing compliance errors.
- **Collaborated cross-functionally** with avionics, UI, and DevOps teams to migrate repositories from JIRA to GitLab, streamline documentation, and integrate testing infrastructure for over 20 simulation components.

Embedded Systems Software Engineer | February 2023 - February 2024

- **Resolved a critical timing defect** impacting 40% of cockpit controls by engineering a full-stack Java fix in the Virtual Flight Deck (VFD). Improved button input accuracy and reduced response latency from **1.0ms to 0.8ms** for higher simulation fidelity.
- **Led modernization of legacy simulation models** from **Fortran and C++ to Rust and Python**, enhancing system performance, code safety, and maintainability. Rust's memory safety eliminated key crash conditions and improved flight deck input latency **by 20%**.
- **Technologies:** JavaScript/TypeScript, React.js, Node.js, Rust, Java, C/C++, Python, Git, Fortran

Software Engineer Intern at Garmin International | Kansas City, KS

May 2022 - August 2022

- **Integrated Vulkan graphics driver** (IMG-BXS-4-64) into Garmin's in-flight display, resolving 1,500 code coverage issues and achieving FAA certification standards for the new Vulkan-based rendering engine. **Developed a custom graphics simulation environment** aligned with Vulkan API specifications, demonstrating advanced understanding of GPU pipelines and rendering performance.
- **Automated source organization** by creating a Python executable that sorted hundreds of functions per file, saving engineers 50% of manual effort and improving project workflow.
- **Technologies:** C/C++, Python, XML, Visual Studio, Vulkan API, Git

Education

Bachelor of Science in Electrical Engineering | Minor in Computer Science

University of Illinois, Urbana-Champaign - Grainger College of Engineering

Years: 2019 - 2022 | GPA: 3.6

Projects

World Metro Explorer (in progress)

Technologies: TypeScript, React.js, Mapbox GL, GeoJSON

- Building an interactive world mapping app to visualize current and future subway networks.
- Integrating open transit datasets with dynamic city filtering and zoomable map layers to explore urban transit expansion.

Am I a Casual?

Technologies: TypeScript, React.js, Mantine Library

- Designed a daily sports trivia site with rotating questions, answer validation, and real-time scoring.
- Built responsive UI optimized for both desktop and mobile users.

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Technologies: TypeScript, React.js, Chakra Library

- Created a daily journaling web app that lets users form groups, share responses, and track reflection habits together.
- Integrated Firebase authentication and real-time syncing for a social, cross-device experience.

Skills

Programming Languages, Frameworks and Libraries

JavaScript, TypeScript, Python, Java, Rust, C/C++, SQL, Swift, HTML, CSS, Assembly (LC-3), SystemVerilog, React.js, Node.js, Mantine, Chakra UI, Firebase